

Anchor–Direct Infimal Convolution on the Orphan Quotient: Exact Hybrid Completion Duality and Stop Certificates

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Abstract

The fixed-shift completion program contains two formally different sources of information. A direct channel synthesizes corrections in the target entry space; an anchor channel records lower-dimensional or four-corner structure. These channels cannot be combined merely because their labels overlap. One must construct a literal anchor synthesis into the same target.

We study the resulting finite-dimensional convex problem. Let $S_A : X_A \rightarrow K$ and $S_D : X_D \rightarrow K$ be anchor and direct synthesis maps, assume S_D is surjective, and assume the coefficient variables form a verified separable direct sum with additive cost $p_A(x) + p_D(y)$. Write

$$\alpha(v) = \inf_{S_A x = v} p_A(x), \quad \delta(k) = \min_{S_D y = k} p_D(y),$$

and $V_A = \text{Ran } S_A$. The direct cost left after declaring anchor directions free is the orphan quotient gauge

$$\delta_{\text{orph}}(\pi k) = \inf_{v \in V_A} \delta(k - v).$$

If $\mathcal{B}_A, \mathcal{B}_D$ are the polar feasibility sets induced by the two coefficient norms, then

$$\delta_{\text{orph}}(\pi k) = \max_{\lambda \in \mathcal{B}_D, S_A^* \lambda = 0} [\lambda, k]_{\mathbb{R}}.$$

The genuinely hybrid cost is instead

$$h(k) = \min_{v \in V_A} \{\alpha(v) + \delta(k - v)\}.$$

We prove the exact dual intersection formula

$$h(k) = \max_{\lambda \in \mathcal{B}_A \cap \mathcal{B}_D} [\lambda, k]_{\mathbb{R}}$$

and the hierarchy

$$\delta_{\text{orph}}(\pi k) \leq h(k) \leq \delta(k).$$

For a fixed datum,

$$h(k) = \delta(k) \iff N_D(k) \cap \mathcal{B}_A \neq \emptyset,$$

where $N_D(k)$ is the face of direct norming functionals. In particular, a direct norming functional annihilating the anchor range certifies that the anchor channel cannot help. Globally, the anchor channel is redundant exactly when $\alpha(v) \geq \delta(v)$ for all $v \in V_A$.

The results give a precise test for whether two previously separate completion mechanisms yield a real saving, only a quotient floor, or no interaction at all. They do not construct the required physical anchor synthesis or prove that its literal fixed-shift coefficients cross any arithmetic endpoint.

Keywords: infimal convolution; quotient gauge; anchor synthesis; direct completion; norming functional; dual intersection; fixed shift.

Convention. Anchor and direct variables may be combined only after their synthesis maps land in the same literal target space. Every physical specialization keeps one fixed nonzero shift h_0 , complete native keys, actual support, literal coefficients, and global raw atomic normalization.

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1 Introduction

The completion program now has two mature but separate interfaces. Direct primitive-entry completion has an exact quotient gauge and primal–dual certificates [3, 4]; sparse forest components admit additional closed reductions [5]. Separately, anchor or lower-content constructions organize four-corner and provenance information. The next logical question is whether an anchor channel can lower the direct completion cost.

That question is ill posed until both channels land in one target space. An observation map out of the direct target, a shared divisor label, or a four-corner trace is not yet a vector that can be subtracted from a two-corner direct entry. The missing object is a synthesis map

$$S_A : X_A \longrightarrow K$$

into the same K used by the direct synthesis $S_D : X_D \rightarrow K$. Once this map exists, three different values appear:

- (i) the direct cost $\delta(k)$;
- (ii) the direct cost on the quotient $K/\text{Ran } S_A$, where anchor directions are declared free;
- (iii) the hybrid cost $h(k)$, where using an anchor direction incurs its actual coefficient cost.

Confusing (ii) and (iii) can manufacture a saving that is not physically available.

Main results. We compute the duals of all three values. The quotient floor uses direct charges annihilating the anchor range. The hybrid cost uses the intersection of the anchor and direct polar balls. This immediately gives

$$\delta_{\text{orph}} \leq h \leq \delta,$$

but the important content is the equality theory. A fixed datum gets no hybrid improvement exactly when a direct norming functional is also anchor feasible. A norming functional that annihilates the entire anchor range is a particularly strong no-improvement certificate. Globally, the anchor channel is redundant exactly when it is never cheaper than the direct channel on its own range.

Relation to the previous layer. TPC-78 reduces the cost of evaluating a direct component. The present paper changes the admissible synthesis family: it asks whether a second family can share the load. Thus it advances a different gate. Forest closed forms can be inserted into δ , but they do not construct S_A or prove that the intersection of polar balls yields a strict physical gain.

Scope and progress label. All theorems are exact finite-dimensional convex analysis over complex spaces viewed as real spaces. They are L0 convex-algebraic results that advance the L1 interface. No literal arithmetic construction of S_A , no carrier-density bound, and no fixed- h_0 endpoint saving is assumed or claimed.

Organization. Sections 2 and 3 define the two synthesis gauges. Section 4 derives the quotient floor. Section 5 proves hybrid duality. Section 6 gives fixed-datum and global equality tests. Section 7 gives matched certificates. Section 8 separates observation from synthesis and quotient from hybrid cost. Sections 9 and 10 state the literal transfer requirements and endpoint stop rule.

2 Common-target synthesis model

Let K, X_A, X_D be finite-dimensional real or complex vector spaces; complex spaces are regarded as real vector spaces when taking duals. Pair each space with its real dual using $[\lambda, k]_{\mathbb{R}}$; after choosing Hermitian coordinates in the complex case this is $\text{Re}\langle \lambda, k \rangle$.

Let

$$S_A : X_A \rightarrow K, \quad S_D : X_D \rightarrow K$$

be real-linear maps. Assume S_D is surjective. Let p_A, p_D be norms on the coefficient spaces, and write p_A°, p_D° for their polar norms. The real adjoints are defined by

$$[S_A^* \lambda, x]_{\mathbb{R}} = [\lambda, S_A x]_{\mathbb{R}}, \quad [S_D^* \lambda, y]_{\mathbb{R}} = [\lambda, S_D y]_{\mathbb{R}}.$$

Definition 2.1 (Polar feasibility sets). Define

$$\mathcal{B}_A = \{\lambda \in K^* : p_A^\circ(S_A^* \lambda) \leq 1\}, \quad (1)$$

$$\mathcal{B}_D = \{\lambda \in K^* : p_D^\circ(S_D^* \lambda) \leq 1\}. \quad (2)$$

Because S_D^* is injective, $\mathcal{B}_D$ is compact. The set $\mathcal{B}_A$ may be unbounded along the annihilator of $\text{Ran } S_A$, but $\mathcal{B}_A \cap \mathcal{B}_D$ is compact.

Definition 2.2 (Anchor range and orphan quotient). Let

$$V_A = \text{Ran } S_A, \quad \pi : K \rightarrow Q_{\text{orph}} := K/V_A$$

be the anchor-visible subspace and its quotient map. A class πk is called the *orphan part* of k : it is the part no anchor synthesis can change.

The term “orphan” is algebraic. It does not assert that a physical carrier is absent; it says only that the chosen literal map S_A does not synthesize that quotient direction.

Remark 2.3 (Observation is not synthesis). A map $T : K \rightarrow Y$ measures or aggregates a target. It does not define a map $Y \rightarrow K$. A right inverse, section, or reconstruction operator must be specified and proved compatible with the physical coefficients before its range may be called V_A .

3 The pure anchor and direct gauges

Definition 3.1 (Pure synthesis gauges). For $v \in K$ and $k \in K$, set

$$\alpha(v) = \inf\{p_A(x) : S_A x = v\}, \quad (3)$$

$$\delta(k) = \min\{p_D(y) : S_D y = k\}. \quad (4)$$

By convention $\alpha(v) = +\infty$ when $v \notin V_A$.

Finite dimensionality gives attainment in both definitions whenever the constraint is feasible. The restriction of α to V_A is a norm, and δ is a norm on K .

Proposition 3.2 (Pure dual formulas). For $v \in V_A$ and $k \in K$,

$$\alpha(v) = \max_{\lambda \in \mathcal{B}_A} [\lambda, v]_{\mathbb{R}}, \quad (5)$$

$$\delta(k) = \max_{\lambda \in \mathcal{B}_D} [\lambda, k]_{\mathbb{R}}. \quad (6)$$

In (5), the objective is constant along annihilator directions, so the maximum is finite and attained after restriction to V_A^* .

Proof. Each gauge is the quotient norm induced by a surjective map onto its range. The polar of a quotient norm consists exactly of those functionals whose pullback has polar norm at most one. This gives (1)–(2) and the displayed support formulas. Equivalently, they follow from finite-dimensional Fenchel duality [1]. $\square$

Definition 3.3 (Direct norming face). For $k \in K$, define

$$N_D(k) = \{\lambda \in \mathcal{B}_D : [\lambda, k]_{\mathbb{R}} = \delta(k)\}.$$

The set $N_D(k)$ is a nonempty compact exposed face of $\mathcal{B}_D$. At $k = 0$, it is all of $\mathcal{B}_D$.

Remark 3.4 (Coefficient cost is part of the model). Replacing p_A or p_D by an unnormalized count changes its polar ball and therefore changes every equality test below. In a physical specialization the coefficient norms must be those induced by the same global raw atomic normalization used in the target estimate.

4 The free orphan quotient floor

Before charging for anchor coefficients, declare all directions in V_A free.

Definition 4.1 (Orphan quotient gauge). For $\pi k \in Q_{\text{orph}}$, define

$$\delta_{\text{orph}}(\pi k) = \inf_{v \in V_A} \delta(k - v). \quad (7)$$

The value is independent of the representative k and is the quotient norm induced by δ .

Theorem 4.2 (Exact orphan dual). *For every $k \in K$,*

$$\delta_{\text{orph}}(\pi k) = \max \{ [\lambda, k]_{\mathbb{R}} : \lambda \in \mathcal{B}_D, S_A^* \lambda = 0 \}. \quad (8)$$

Both extrema are attained.

Proof. The annihilator of $V_A = \text{Ran } S_A$ is

$$V_A^\perp = \{ \lambda : S_A^* \lambda = 0 \}.$$

The dual unit ball of a quotient norm is the intersection of the original dual unit ball with the annihilator of the quotient subspace. Applying this to δ proves (8). Direct finite-dimensional minimax duality gives the same result: the free variable $v \in V_A$ forces its dual coefficient to vanish on V_A . $\square$

Corollary 4.3 (Exact orphan zero test).

$$\delta_{\text{orph}}(\pi k) = 0 \iff k \in V_A.$$

Proof. It is a norm on K/V_A . $\square$

Remark 4.4 (A floor, not an attainable hybrid price). The minimization in (7) charges nothing for v . It is a lower floor on any model that assigns a nonnegative cost to anchor synthesis. It is not the actual joint cost unless the required anchor directions are free in that model.

5 Hybrid infimal convolution and dual intersection

Definition 5.1 (Hybrid anchor–direct gauge). For $k \in K$, define

$$h(k) = \min_{v \in V_A} \{ \alpha(v) + \delta(k - v) \} \quad (9)$$

$$= \min_{\substack{x \in X_A, y \in X_D \\ S_A x + S_D y = k}} \{ p_A(x) + p_D(y) \}. \quad (10)$$

The two formulas agree by the definition of the quotient coefficient norms. Since S_D is surjective, h is a finite norm on K .

Theorem 5.2 (Exact hybrid duality). *For every $k \in K$,*

$$h(k) = \max_{\lambda \in \mathcal{B}_A \cap \mathcal{B}_D} [\lambda, k]_{\mathbb{R}}. \quad (11)$$

The primal and dual extrema are attained.

Proof. Regard $X_A \oplus X_D$ as a normed space with

$$p(x, y) = p_A(x) + p_D(y)$$

and synthesis $S(x, y) = S_A x + S_D y$. Its polar norm is

$$p^\circ(\xi, \eta) = \max \{ p_A^\circ(\xi), p_D^\circ(\eta) \}.$$

The quotient-norm dual formula therefore imposes both

$$p_A^\circ(S_A^* \lambda) \leq 1, \quad p_D^\circ(S_D^* \lambda) \leq 1.$$

These are precisely $\lambda \in \mathcal{B}_A \cap \mathcal{B}_D$. Finite-dimensional quotient-norm duality gives attainment [1, 2]. $\square$

Corollary 5.3 (Three-level hierarchy). *For every $k \in K$,*

$$\delta_{\text{orph}}(\pi k) \leq h(k) \leq \delta(k). \quad (12)$$

Proof. If $S_A^* \lambda = 0$, then $p_A^\circ(S_A^* \lambda) = 0$, so the orphan dual feasible set is contained in $\mathcal{B}_A \cap \mathcal{B}_D$. This gives the first inequality. The second follows either from $\mathcal{B}_A \cap \mathcal{B}_D \subseteq \mathcal{B}_D$, or by taking $v = 0$ in (9). $\square$

The two inequalities measure different gaps. The upper gap $\delta - h$ is an actual saving produced by a costed anchor channel. The lower gap $h - \delta_{\text{orph}}$ is the price of realizing the chosen anchor-visible correction.

6 Exact equality and strict-improvement tests

Theorem 6.1 (Fixed-datum no-improvement certificate). *For every $k \in K$,*

$$h(k) = \delta(k) \iff N_D(k) \cap \mathcal{B}_A \neq \emptyset. \quad (13)$$

Equivalently, the anchor channel gives a strict saving exactly when every direct norming functional violates the anchor polar constraint.

Proof. The hybrid feasible set is the compact subset $\mathcal{B}_A \cap \mathcal{B}_D$ of the direct feasible set $\mathcal{B}_D$. Their support functions agree at k precisely when a direct maximizer belongs to the subset. This is exactly (13). $\square$

Corollary 6.2 (Orphan norming obstruction). *If there exists $\lambda \in N_D(k)$ with $S_A^* \lambda = 0$, then*

$$\delta_{\text{orph}}(\pi k) = h(k) = \delta(k).$$

Thus no anchor synthesis, regardless of its coefficient price, can improve the direct cost at k .

Proof. Such a λ is feasible for the orphan dual and attains $\delta(k)$. Combine (12) with the reverse lower bound supplied by λ . $\square$

Theorem 6.3 (Global redundancy criterion). *The following are equivalent:*

- (i) $h(k) = \delta(k)$ for every $k \in K$;
- (ii) $\alpha(v) \geq \delta(v)$ for every $v \in V_A$;
- (iii) $\mathcal{B}_D \subseteq \mathcal{B}_A$.

Proof. Assume (ii). For every $v \in V_A$,

$$\alpha(v) + \delta(k - v) \geq \delta(v) + \delta(k - v) \geq \delta(k).$$

Taking the infimum and using $h \leq \delta$ gives (i). Conversely, for $v \in V_A$,

$$\delta(v) = h(v) \leq \alpha(v) + \delta(0) = \alpha(v),$$

so (i) implies (ii).

By the pure support formulas, $\alpha \geq \delta$ on V_A is equivalent to every $\lambda \in \mathcal{B}_D$ restricting to an α -dual feasible functional on V_A , which is exactly $\mathcal{B}_D \subseteq \mathcal{B}_A$: support-function domination first gives inclusion after restriction to V_A^* , and $\mathcal{B}_A$ is saturated along $V_A^\perp$, so the inclusion lifts back to K^* . Alternatively, (iii) makes the two dual support sets $\mathcal{B}_A \cap \mathcal{B}_D$ and $\mathcal{B}_D$ identical. $\square$

Proposition 6.4 (When the quotient floor is attained). *Let*

$$N_H(k) = \{\lambda \in \mathcal{B}_A \cap \mathcal{B}_D : [\lambda, k]_{\mathbb{R}} = h(k)\}.$$

Then

$$h(k) = \delta_{\text{orph}}(\pi k) \iff N_H(k) \cap \text{Ker } S_A^* \neq \emptyset.$$

Proof. The orphan dual feasible set is precisely $(\mathcal{B}_A \cap \mathcal{B}_D) \cap \text{Ker } S_A^*$. Equality of the two support values is therefore equivalent to the existence of a hybrid maximizer in that subset. $\square$

7 Matched certificates and perturbation stability

Theorem 7.1 (Decomposed primal–dual gap). *Let $x \in X_A$, $y \in X_D$ satisfy*

$$S_A x + S_D y = k,$$

and let $\lambda \in \mathcal{B}_A \cap \mathcal{B}_D$. Define

$$U = p_A(x) + p_D(y), \quad L = [\lambda, k]_{\mathbb{R}}.$$

Then

$$L \leq h(k) \leq U \tag{14}$$

and the gap decomposes exactly as

$$\begin{aligned} U - L &= (p_A(x) - [S_A^* \lambda, x]_{\mathbb{R}}) \\ &\quad + (p_D(y) - [S_D^* \lambda, y]_{\mathbb{R}}). \end{aligned} \tag{15}$$

Both bracketed terms are nonnegative. Hence $U = L$ is a complete, separately auditable optimality certificate.

Proof. Polar feasibility gives

$$[S_A^* \lambda, x]_{\mathbb{R}} \leq p_A(x), \quad [S_D^* \lambda, y]_{\mathbb{R}} \leq p_D(y).$$

Using the synthesis equality gives

$$L = [\lambda, S_A x + S_D y]_{\mathbb{R}}.$$

Subtracting proves (15); weak duality and primal feasibility give (14). $\square$

Corollary 7.2 (Phase saturation conditions). *If p_A, p_D are coordinate ℓ^1 norms, a zero gap requires coordinatewise saturation in both channels: every nonzero coefficient must align with a unit-modulus coordinate of the pulled-back dual charge. Vanishing stationarity residual alone is not a lower certificate.*

Proposition 7.3 (Lipschitz stability). *For all $k, k' \in K$,*

$$|h(k) - h(k')| \leq h(k - k') \leq \delta(k - k').$$

Likewise,

$$|\delta_{\text{orph}}(\pi k) - \delta_{\text{orph}}(\pi k')| \leq \delta_{\text{orph}}(\pi(k - k')).$$

Proof. Both h and δ_{orph} are norms on their respective spaces, so the reverse triangle inequality applies. The second inequality for h is [corollary 5.3](#). $\square$

Remark 7.4 (Approximate computations). A numerical primal point always gives an upper bound if the synthesis equation is enforced exactly or its residual is charged explicitly. A numerical dual point gives a lower bound only after both polar constraints have been verified. Projecting a nearly feasible charge or reporting its violation is preferable to silently treating a tolerance as feasibility.

8 Countermodels: distinctions that cannot be skipped

Example 8.1 (Strict three-level hierarchy). *Let $K = \mathbb{R}^2$, $S_D = \text{id}$, $p_D(y) = \|y\|_1$, and*

$$S_A a = (a, 0), \quad p_A(a) = c|a|, \quad 0 < c < 1.$$

Then

$$\delta(k) = |k_1| + |k_2|, \quad \delta_{\text{orph}}(\pi k) = |k_2|,$$

whereas

$$h(k) = c|k_1| + |k_2|.$$

For $k_1 k_2 \neq 0$, both inequalities in (12) are strict. The quotient floor is not the hybrid price, but the anchor channel does give a genuine saving.

Example 8.2 (An expensive anchor gives no saving). *Keep the same maps but take $c \geq 1$. Then*

$$h(k) = \delta(k) = |k_1| + |k_2|, \quad \delta_{\text{orph}}(\pi k) = |k_2|.$$

The anchor range is nontrivial and the free quotient floor is smaller, yet the costed anchor is globally redundant. This is [theorem 6.3](#) in its simplest form.

Example 8.3 (Full anchor range does not mean zero cost). *Let $K = \mathbb{R}$, $S_A = S_D = \text{id}$, $\delta(k) = |k|$, and $\alpha(k) = 2|k|$. Then*

$$Q_{\text{orph}} = \{0\}, \quad \delta_{\text{orph}} = 0, \quad h(k) = \delta(k) = |k|.$$

Declaring the full target anchor-visible erases the quotient but does not make anchor coefficients free.

Example 8.4 (Observation does not determine synthesis). *Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ be $T(x_1, x_2) = x_1 + x_2$. The two sections*

$$R_1(a) = (a, 0), \quad R_2(a) = (0, a)$$

both satisfy $TR_\ell = \text{id}$, but they produce different anchor ranges. For example, with $\delta(x_1, x_2) = |x_1| + 2|x_2|$ and $p_A(a) = |a|$, the two induced hybrid gauges are

$$h_1(x_1, x_2) = |x_1| + 2|x_2|, \quad h_2(x_1, x_2) = |x_1| + |x_2|.$$

Therefore the observation $T(k)$ does not specify which target entry can be cancelled. A chosen reconstruction must be part of the theorem, not a post hoc identification.

Example 8.5 (A quotient gauge is not orthogonal projection). *Let*

$$\delta(x, y) = |x| + 2|y|, \quad V_A = \text{span}\{(1, 1)\}, \quad k = (1, 0).$$

Then

$$\inf_{t \in \mathbb{R}} \delta((1, 0) - t(1, 1)) = 1,$$

attained at $t = 0$. Euclidean orthogonal projection uses $t = 1/2$ and leaves direct cost $3/2$. Quotient minimization is governed by the literal direct gauge, not an auxiliary Hilbert metric.

Remark 8.6 (Shared labels do not imply independent channels). If anchor and direct columns share native coefficient variables, the coefficient space is not the direct sum used in (10). One must retain the shared variable and use its actual joint norm. Duplicating it into two independently optimized copies can only decrease the computed cost and may create a false saving.

9 Requirements for a literal fixed-shift transfer

Fix one nonzero physical shift h_0 . A valid specialization at scale X must archive the following common-target diagram:

$$\begin{array}{ccc} X_{A,X} & \xrightarrow{S_{A,X}} & K_X \\ & \uparrow S_{D,X} & \\ & X_{D,X} & \end{array}$$

The arrows must be literal coefficient assemblies. A trace, restriction, aggregate, or four-corner observation into another space does not fill the top arrow.

Definition 9.1 (Physical hybrid packet). A physical packet to which the intersection formula applies consists of:

- (H1) one fixed h_0 and one target key set K_X ;
- (H2) literal maps $S_{A,X}, S_{D,X}$, including all provenance and content lifts before exact summation, with $S_{D,X}$ surjective onto K_X ;
- (H3) coefficient norms $p_{A,X}, p_{D,X}$ derived from the same raw atomic measure;

- (H4) the datum k_X in the same units and orientation as both synthesis maps;
- (H5) a verified separable coefficient space $X_{A,X} \oplus X_{D,X}$ with cost $p_{A,X} + p_{D,X}$;
- (H6) primal coefficients and a dual charge with both polar constraints verified;
- (H7) a separately proved arithmetic inequality connecting the physical quantity of interest to δ_X, h_X , or their difference.

If anchor and direct channels share or couple native variables, the packet is not separable and $\mathcal{B}_A \cap \mathcal{B}_D$ need not be its dual ball. One must instead construct the literal joint synthesis, joint coefficient norm, and its single polar ball; none of the intersection formulas may be applied until an exact separable reparameterization is proved.

Proposition 9.2 (Common-target necessity). *Suppose anchor data are available only through an observation $T_X : K_X \rightarrow Y_X$. The values $T_X k_X$, the direct gauge δ_X , and a norm on Y_X do not in general determine a hybrid value on K_X . A synthesis $R_X : Y_X \rightarrow K_X$, or a direct coefficient map from anchor variables into K_X , is necessary.*

Proof. [Example 8.4](#) gives two sections of the same observation with distinct target ranges and therefore potentially distinct hybrid gauges. The input data listed in the statement cannot distinguish them. $\square$

Remark 9.3 (Majorization direction). Convex duality computes a finite value; it does not decide whether that value upper-bounds, lower-bounds, or equals a physical bad-set mass. The direction and every scale factor must be proved from the literal arithmetic assembly. A saving in h_X is not an endpoint saving until this bridge is supplied.

The minimal reproducible census records, for each X , the dimensions and hashes of all native arrays, exact output keys, ranks and actual supports of both synthesis maps, the values

$$\delta_{\text{orph},X}(\pi k_X), \quad h_X(k_X), \quad \delta_X(k_X),$$

and their globally normalized gaps. Numerical feasibility tolerances must be reported separately from exact support and exact adjoint constraints.

10 Claim boundary, stop certificates, and next gate

What is proved

For any finite literal, separable, surjective-direct common-target packet, this paper proves:

- (C1) exact direct, orphan, and hybrid dual formulas;
- (C2) the hierarchy $\delta_{\text{orph}} \leq h \leq \delta$;
- (C3) a necessary-and-sufficient fixed-datum no-improvement test;
- (C4) an orphan norming functional as a strong sufficient local stop certificate;
- (C5) a necessary-and-sufficient global redundancy test;
- (C6) matched decomposed primal–dual gaps.

What is not proved

We do not prove that the previously available anchor observables admit a canonical synthesis into the direct two-corner target. We do not prove that the literal fixed- h_0 anchor polar ball removes every direct norming functional, that strict saving occurs on a positive density of carriers, or that the inherited total exponent-loss ledger closes strictly below $1/400$. We do not convert a support-only or averaged-family result into fixed- h_0 parity progress.

Level	Status
L0: model algebra	Exact for finite real-linear common-target synthesis.
L1: interface	Advanced by exact quotient/hybrid separation and equality certificates.
L2: arithmetic	Open; the literal anchor synthesis and fixed- h_0 transfer estimate are not supplied.

Branch stop certificates

At a fixed datum, a verified witness

$$\lambda \in N_D(k) \cap \mathcal{B}_A$$

rigorously proves no anchor improvement. A primal–dual hybrid pair alone only brackets $h(k)$. A certified strict saving requires, for example, a direct lower bound L_D and a hybrid upper bound U_H with $L_D > U_H$, or exact evaluation of both values. Solver failure proves neither alternative. Globally, $\alpha \geq \delta$ on V_A stops the anchor branch entirely. If a physical common-target map is not yet constructed, the gate remains open; only a proof of incompatibility supports a negative transfer theorem.

Endpoint loss ledger and next gate

The inherited fixed-power ledger allows total exponent loss strictly smaller than $1/400$ and stops at loss at least $1/400$, including equality. The exact finite-dimensional identities here add no exponent loss by themselves, but a physical synthesis dictionary, carrier restriction, and reassembly theorem may do so. A strict hybrid gauge gap is not an exponent gain until a separate transfer theorem proves its direction, scale, and common global normalization.

The next structural question is orthogonal: when a Hermitian physical matrix is bounded entrywise by a nonnegative Perron majorant, exactly when do complex phases allow the Perron bound to be saturated? That phase-consistency problem requires cycle holonomy and must be kept separate from the incidence and orphan-quotient graphs studied here.

Conclusion

The anchor channel has now been placed on an exact decision tree:

$$\begin{array}{c}
\text{construct a literal common-target synthesis} \\
\downarrow \\
\delta_{\text{orph}} \leq h \leq \delta \\
\downarrow \\
\left\{ \begin{array}{ll} \text{norming charge survives} & \Rightarrow \text{no improvement,} \\ \text{dual face is cut} & \Rightarrow \text{certifiable hybrid saving.} \end{array} \right.
\end{array}$$

This identifies both a usable route and a rigorous stopping mechanism. The remaining difficulty is no longer hidden in the convex program: it is the construction and arithmetic prevalence of the physical channel.

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